

Hi Zahid,

Thank you for your constructive feedback. I particularly appreciate your comments on the “Weakness–Trigger–Impact” model. Your observation that this approach captures the interaction between user behaviour and technical vulnerabilities aligns closely with the broader emphasis in secure software development on socio-technical risk (Olmsted, 2024).

Your suggestion to expand on the limitations of BitLocker is a good one. Although BitLocker benefits from strong integration with the Windows ecosystem and the Trusted Platform Module (TPM), it is not without weaknesses. Its reliance on platform trust introduces potential risks in the presence of firmware or hardware-based attacks, while its proprietary nature limits independent verification of the implementation. This creates a different type of assurance gap compared to open-source alternatives, where transparency enables external scrutiny.

I also agree that elaborating on mitigation strategies would strengthen the analysis. In the case of VeraCrypt, enforcing strong password policies, encouraging passphrase-based authentication, and promoting secure keyfile management can significantly reduce the likelihood of misconfiguration. More broadly, this reflects a central principle of the secure software development lifecycle: controls must not only be embedded in system design, but also in user guidance and operational practice. Without this, even well-designed cryptographic systems remain vulnerable to misuse.

Your point therefore marks an important extension of the original argument. The comparison is not simply between tools, but between different security models: one prioritises usability and controlled risk, while the other prioritises transparency and user control. The appropriate choice therefore depends less on any presumed superiority, and more on the user’s capacity and context in managing the associated risks.

Thank you again for your valuable suggestions; they are very helpful in refining my summary post for the e-portfolio.

Kind regards, Victor